MIL-C-17858C(AS)
1 JUNE 1966
SUPERSEDING
MIL-C-17858B(WEP)
15 September 1964

MILITARY SPECIFICATION

COMPASS SYSTEM: MAGNETIC SLAVED, DIRECTIONAL GYRO STABILIZED, 4°/HR, TYPE MA-1

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design requirements and performance requirements for a 4 degree per hour, magnetic slaved, directional gyro stabilized compass system for military aircraft.

1.2 Classification - The compass system shall consist of the items listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.10 of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

* 3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(a) Parts and materials requirements shall conform to MIL-E-5400.

FSC 6605

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(b) Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4.)

(c) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

* 3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

* 3.2.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied as outlined in MIL-E-4682.

* 3.3 Design and Construction - The compass system shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein. The compass system shall include components and devices such as shock mounts, junction box, etc., as necessary to satisfy the test requirements specified in Section 4. The compass system shall be operable with a remote compass transmitter. The compass system shall provide accurate stabilized signals of the heading of an aircraft through 360 degrees in azimuth. The azimuth signal shall be presented by two high precision transmitting synchros as outputs for indicators and accessory equipment such as OMNI range, radar stabilization, etc. The operation of the system shall be such that it shall function as a compass controlled directional gyro or a free directional gyro with correction for the effect of the earth's rotation. The random drift of the entire system shall not exceed 4 degrees per hour when operated as a free gyro compass system with latitude control. The compass system shall include a synchronizing indicator which will provide visual indication of misalignment between the compass and directional gyro. It is required that the system be flexible enough to incorporate a 24 point compass signal deviation compensator or equivalent in the transmitter - gyro section of the system. The compass system shall be so constructed that no parts will work loose in service and will withstand the normal shocks, vibrations and such other conditions as are incident to service without failure, shipping, storage, and installation.

3.3.1 Total Weight - The total weight of the compass system, excluding cables, shall not exceed 17 pounds.

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* 3.3.2 Reliability -

3.3.2.1 Operational Stability - The compass system shall operate with satisfactory performance, continuously or intermittently for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Longevity - The compass system shall have a total operating life of 1000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failures (MTBF) - For the system test program under 4.4.3, the mean (operating) time between failures shall be 800 hours.

3.3.2.4 Time Totalizing Meter - The system shall contain time totalizing meters (10,000 hours) in accordance with MIL-M-7793. The reading of the meters shall not require unsealing of the components. The time totalizers shall be installed within the present envelope of the components. Meters shall be included in the following components:

Component	Type of Meter
Directional Gyro	MS17322
Amplifier	MS17322

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The compass system shall provide for the use of cables and connectors in accordance with MIL-E-5400.

* 3.3.3.2 Interconnection Cabling - The compass system shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the compass system.

* 3.3.4 Interchangeability of Reordered Equipment - For re-ordered equipment, interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment supplied or designated by the procuring activity.

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3.3.5 Interference Control - The generation of radio interference by the compass system and the vulnerability of the compass system to radio interference shall be controlled within the limits of MIL-I-6181.

* 3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

* 3.3.7 Nameplates - Nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 ±5° C)
Altitude	Normal ground
Pressure	Normal atmospheric (approximately 29.92 inches Hg)
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115/200 ±5 VAC, 4 wire grounded, 400 ±10 cps, 28 ±1 VDC, grounded negative

* 3.3.9 Service Conditions - The compass system shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein:

3.3.9.1 Vibration - The compass system shall operate satisfactorily when subjected to the vibration requirements of Curve IV of MIL-E-5400, except as specified herein.

3.3.10 Warm-Up Time - The time required for the compass system to warm-up prior to operation shall be kept to a minimum and shall not exceed 3 minutes under standard conditions and 10 minutes at extreme service conditions.

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3.3.11 Primary Input Power Requirements - The system shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(a) AC Power (Three-Phase) 115/200V, 4 wire grounded neutral 400 cps, Category "B" 300 VA

(b) DC Power 28V, grounded negative, Category "B" 40 Watts.

3.3.11.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the compass system.

3.3.12 Standby Provisions - The compass system shall provide for a standby position whereby rated voltage shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the compass system.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5 Detail Requirements -

3.5.1 Directional Gyro - The directional gyro shall be in accordance with MS90317 or MS90320, as applicable. The directional gyro shall be electrically driven and shall be of the non-tumbling type. It shall contain an erection device and one or more synchro transmitters. The gyro shall provide the directional reference to the compass system on one of the synchro transmitters.

3.5.1.1 Erection Device - The erection device shall be capable of maintaining the gyro in its erected position with a leveling rate of 2 to 4 degrees per minute.

3.5.2 Controller - The controller shall be in accordance with MS90316 or MS90319, as applicable. The controller shall include the controls and indicating devices of the compass system conforming to the requirements

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of MIL-C-6781, MIL-P-007788 and MIL-F-25173. The controller shall provide means to select either of the following types of system operation; compass controlled direction gyro, or free directional gyro with correction for the effect of earth's rotation. The controller shall also provide a manual setting of the system to any heading and a synchronizing indicator.

3.5.2.1 Synchronizing Indicator - The arrangement of the synchronizing indicator shall be such that when the gyro output signal is less than the compass transmitter signal, the pointer of the synchronizing indicator shall be to the left of its center (synchronized) position. When the pointer is to the right of its center position, the gyro output signal shall be greater than the compass transmitter signal. Maximum deflection of the pointer shall correspond to a signal difference of 6 degrees.

3.5.2.2 Synchronizing Knob - The controller shall include a synchronizing knob which shall change the output signal at a rate of 720 degrees per minute. Clockwise rotation or deflection to the right shall cause an increase of the output signal. The synchronizing knob shall be a "pull to set" or similar type, such that the output signal cannot be inadvertently changed.

3.5.2.3 Latitude Control - The controller shall include a latitude control to set in the appropriate north or south latitude to increase or decrease the output signal to counteract the effect of the earth's rotation when the system is operated as a free gyro.

3.5.3 Amplifier - The amplifier shall be in accordance with MS90315 or MS90318, as applicable. The compass system shall include an electronic amplifier unit containing amplifiers as required to slave the gyro and position the azimuth control transformer and the output synchros.

3.5.3.1 Output Synchros - The nominal electrical characteristics of the output synchros shall be as specified in Table I.

TABLE I

NOMINAL ELECTRICAL CHARACTERISTICS

NO LOAD	LOADED (1 CONTROL TRANSFORMER)
Input Voltage 26V, 400 cycles	Input Voltage 26V, 400 Cycles
Input Current .225 Amp.	Input Current .260 Amp.
Input Power 1.25 Watts	Input Power 1.5 Watts

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TABLE I (Continued)

NO LOAD	LOADED (1 CONTROL TRANSFORMER)
Input Impedance 25+J 115 Ohms	Input Impedance 22+J 100 Ohms
Output Voltage (Max) 11.8 Volts (Line to Line)	Output Voltage (Max) 20 Volts (Across C.T. Rotor)
D. C. Rotor Resistance 9.5 Ohms	Voltage Phase Shift 13 Deg.
D. C. Stator Resistance 3.5 Ohms (Line to Line)	Sensitivity 350 MV/Deg.

3.5.4 General Requirements -

3.5.4.1 Auto-pilot Output - The compass system shall include a synchro having electrical characteristics conforming to Table I for operation of an automatic pilot.

* 3.5.4.2 Automatic Cutout (System B Only) - The compass system shall be capable of accepting an automatic cutout of the compass control and gyro erection during turns.

3.5.5 Case - The outline dimensions of each component of the compass system shall conform to the Military Standards specified in 6.10. The case shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service.

* 3.5.6 Lighting - The design of the compass controller panel shall incorporate lighting in accordance with MIL-P-007788C(WEP).

3.5.7 Electric Connectors - The electric connectors used shall conform to MIL-C-5015. The type connector used shall be as shown on the applicable component military standard specified in 6.10. The connector code

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number markings as specified in MIL-E-5400 shall be legibly and permanently applied to the case or connector of the applicable component.

* 3.5.8 Wiring - The inter-connect wiring shall be as shown in Figure 1, (System A) or Figure 2 (System B). Internal wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering, where practicable.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

* 4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction Tests

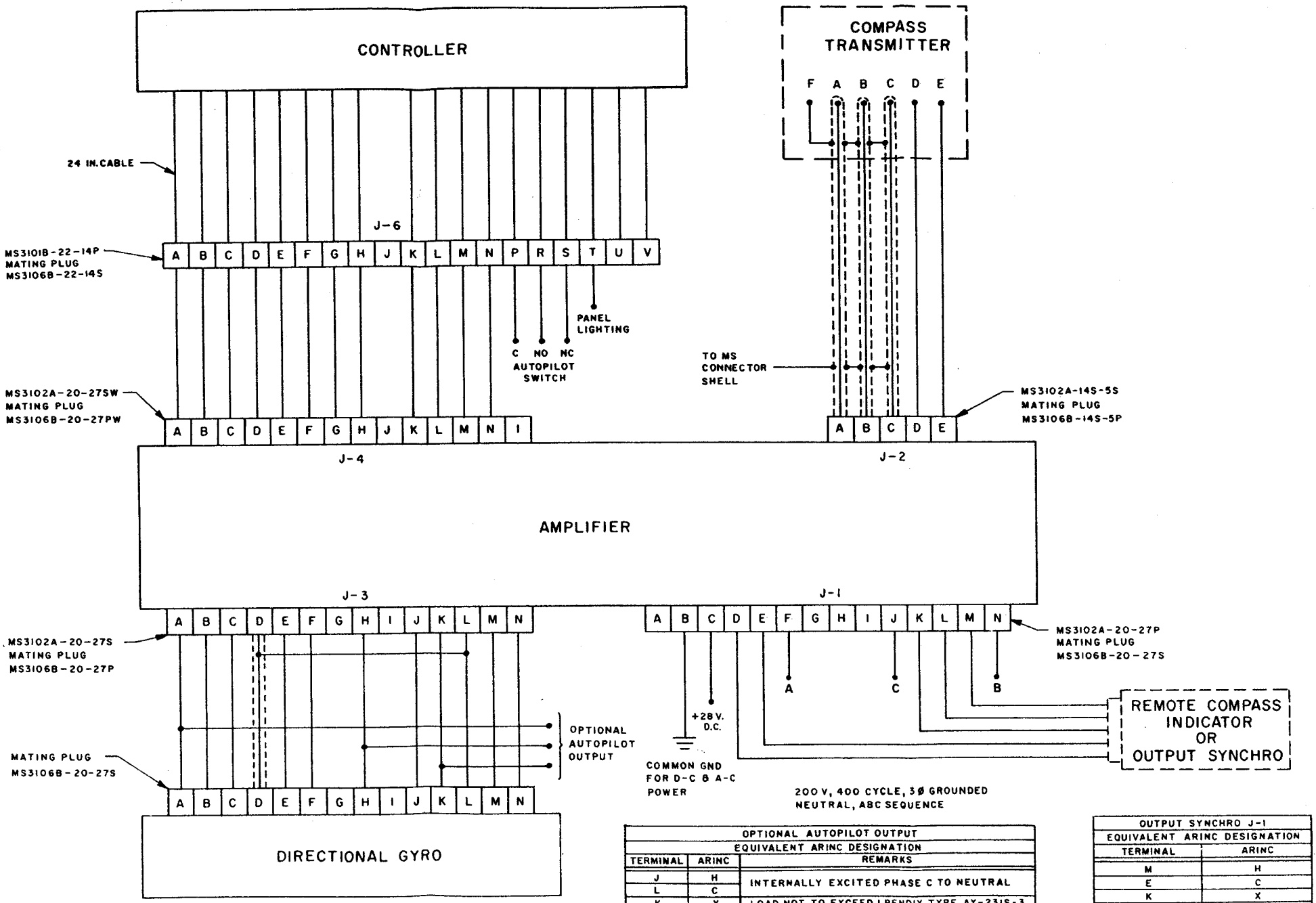
(b) Initial Production Tests

(c) Acceptance Tests

(d) Longevity Tests

* 4.2 Preproduction Tests - Preproduction tests shall be made on a compass system representative of the production compass systems to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

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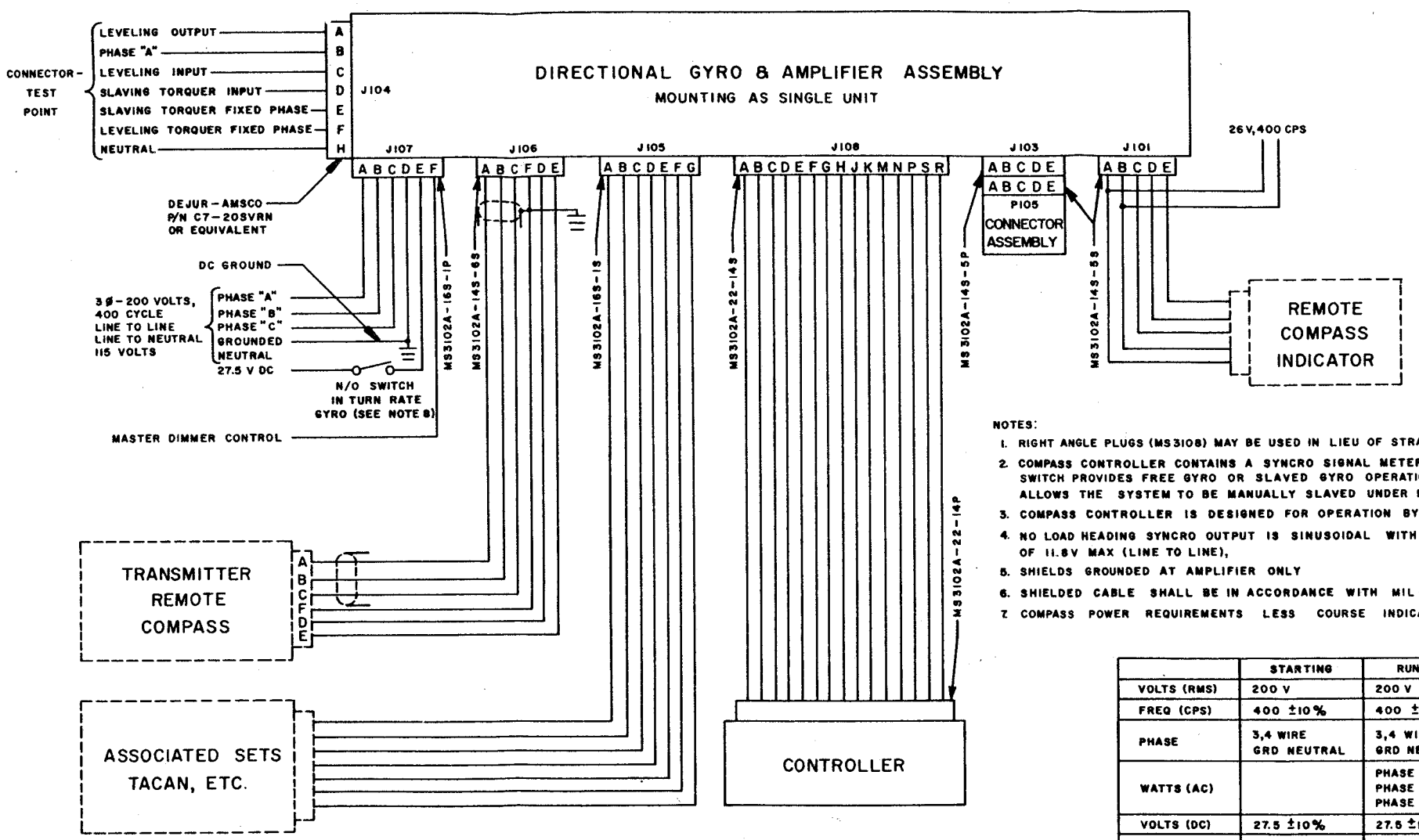


OPTIONAL AUTOPILOT OUTPUT		
EQUIVALENT ARINC DESIGNATION		
TERMINAL	ARINC	REMARKS
J	H	INTERNALLY EXCITED PHASE C TO NEUTRAL
L	C	
K	X	LOAD NOT TO EXCEED 1 BENDIX TYPE AY-231S-3
H	Y	DIFFERENTIAL OR EQUIV FEEDING 1 BENDIX TYPE
D	Z	AY-208S-5 CONTROL TRANS OR EQUIV FEEDING
		A MIN LOAD OF 50,000 Ω IMPEDANCE.

OUTPUT SYNCHRO J-1	
EQUIVALENT ARINC DESIGNATION	
TERMINAL	ARINC
M	H
E	C
K	X
L	Y
D	Z
26 V, 400 CYCLE EXCITATION TO BE SUPPLIED BY BUWEPS	

*Figure 1. Inter-Connect Wiring Diagram (System A)

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NOTES:

1. RIGHT ANGLE PLUGS (MS3108) MAY BE USED IN LIEU OF STRAIGHT PLUGS (MS3106).
2. COMPASS CONTROLLER CONTAINS A SYNCRO SIGNAL METER (ANNUNCATOR). A TOGGLE SWITCH PROVIDES FREE GYRO OR SLAVED GYRO OPERATION AND ROTARY SWITCH ALLOWS THE SYSTEM TO BE MANUALLY SLAVED UNDER FREE GYRO OPERATION.
3. COMPASS CONTROLLER IS DESIGNED FOR OPERATION BY THE PILOT'S RIGHT HAND.
4. NO LOAD HEADING SYNCRO OUTPUT IS SINUSOIDAL WITH A MAX NORMAL VOLTAGE OF 11.8V MAX (LINE TO LINE),
5. SHIELDS GROUNDED AT AMPLIFIER ONLY
6. SHIELDED CABLE SHALL BE IN ACCORDANCE WITH MIL-C-7078, TYPE II.
7. COMPASS POWER REQUIREMENTS LESS COURSE INDICATORS ARE AS FOLLOWS:

	STARTING	RUNNING
VOLTS (RMS)	200 V	200 V
FREQ (CPS)	400 ±10%	400 ±10%
PHASE	3,4 WIRE GRD NEUTRAL	3,4 WIRE GRD NEUTRAL
WATTS (AC)		PHASE A 7.5 PHASE B 4.0 PHASE C 6.0
VOLTS (DC)	27.5 ±10%	27.5 ±10%
WATTS (DC)		

8. THE RATE GYRO REQUIRED TO ENERGIZE PIN "E" OF J107 IS NOT PART OF THE MA-1 SYSTEM.
9. PINS "F" & "G" OF J105 GO TO NORMALLY CLOSED SWITCH IN CONTROLLER. SWITCH IS OPENED WHEN NEW HEADING IS SET BY HEADING KNOB. SWITCH IS USED FOR AUTOPILOT DISCONNECT.

* Figure 2. Inter-Connect Wiring Diagram (System B)

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* 4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

* 4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the compass system meets all the requirements of this specification and the contract, and as a minimum shall include the tests listed under Test Methods.

* 4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production compass systems shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of compass systems to be submitted for acceptance. The preproduction sample shall not be considered as one of the compass systems under the contract.

* 4.2.4 Production Equipments - Compass systems supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each compass system shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for compass systems already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

* 4.3 Initial Production Tests - One of the first ten production compass systems shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This compass system shall be selected by the procuring activity after the system has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

* 4.3.1 Scope of Tests - The compass system may be subjected to any and all tests the procuring activity deems necessary to assure that the production compass system is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements. As a minimum, the compass system shall be subjected to the following tests:

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(a) Sampling Tests

(b) Radio Noise Interference (4.8.20)

(c) Automatic Cutout of Compass Control and Gyro Leveling (4.8.21)

(d) Room Temperature - High Altitude (4.8.22)

(e) Low Temperature - High Altitude (4.8.23)

(f) Shock (4.8.24)

(g) Humidity (4.8.25)

(h) Salt Spray (4.8.26)

(i) Fungus Resistance (4.8.27)

* 4.3.2 Accessory Material - In addition to the complete compass system submitted for Initial Production Tests, the contractor shall also submit such accessory material and data as are necessary to test the compass system.

* 4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of compass systems under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on compass systems previously accepted when requested by the procuring activity.

* 4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the compass system shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the compass system for acceptance.

* 4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be

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signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests

(b) Sampling Tests

(c) Reliability Assurance Tests

(d) Special Tests

* 4.4.1 Individual Tests - Each compass system submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each compass system accepted shall have passed the following tests:

Examination of Product (4.8.1)
Nutation (4.8.2)
Anti-Spin Brake (System B only) (4.8.3)
Leveling (4.8.4)
Synchronizing Indicator Deflection (4.8.5)
Operation Selection (4.8.6)
Direction of Rotation (4.8.7)
Compass Control Rate (4.8.8)
System Scale Error (Free Gyro) (4.8.9)
System Scale Error (Slaved Standard and Weak Field) (4.8.10)

* 4.4.2 Sampling Tests - Compass systems selected for sampling tests shall first have passed the individual tests. Compass systems shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Compass Systems Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0 1/
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 100 or fraction thereof

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1/ NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one compass system from the first 10 produced.

* 4.4.2.1 Scope of Tests - As a minimum, each compass system selected for sampling tests shall be subjected to the following tests:

Power Requirement (4.8.11)
Null Shifts in Slaving Amplifier (4.8.12)
Servoed and Non-servoed Heading Synchro Transmitter Operation (4.8.13)
Auto-Pilot Cutout Switch (4.8.14)
Voltage and Frequency Variation (4.8.15)
Low Temperature Operation (4.8.16)
High Temperature Operation (4.8.17)
Rotation of Spin Axis through 360 Degrees (4.8.18)
Vibration (4.8.19)

* 4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by MIL-STD-781. Compass systems selected for reliability assurance tests shall first have passed the individual tests.

* 4.4.3.1 Reliability Qualification Phase - Prior to the acceptance of compass systems under the contract or order, a minimum of three (3) compass systems shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of compass systems to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

* 4.4.3.2 Reliability Sampling Phase Tests - Samples of the compass system shall be tested as outlined in MIL-STD-781, under the Section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase, Test Level F shall be used. The Accept-Reject criteria for Test Plan VI shall be used to determine the length of the tests (until an accept or reject decision is reached).

* 4.4.3.2.1 Lot Size for Sampling Phase - The entire contract quantity (with the exception of those compass systems used in the Qualification Phase) shall be considered one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples shall be selected and placed on test in the same manner as though one month's production was a lot. The samples shall be tested until an accept or reject decision is reached. In the meantime other compass systems constructed shall be shipped. The test results shall be summarized monthly for the procuring

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activity. The procuring activity reserves the right to stop the acceptance of compass systems at any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the compass system, the compass system parts, the compass system quality control, etc., so that the entire contract quantity will show an accept decision.

* 4.4.3.3 Test Details - The test details such as the length of the Test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted to and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests. The following paragraphs shall be considered minimum conditions and shall apply to both the Qualification and Sampling Phases.

* 4.4.3.3.1 Duty Cycle - Each compass system submitted for reliability assurance tests shall be subjected to a series of operating cycles as follows:

(a) Each cycle shall be 3 hours duration.

(b) The compass system shall be electrically interconnected and shall be energized for 2 1/2 hours of each 3 hour cycle.

(c) During the 2 1/2 hour operating period of each cycle, the gyro shall be mounted on a pitch, roll and yaw test apparatus as described in 4.7.6 herein and shall be subjected to a ±1 1/2 degree roll, pitch, and yaw motion.

(d) Seven of every eight cycles to which the compass system is subjected, shall be performed as described in (c) above and the eighth cycle shall be as follows: The compass system shall be energized and subjected to sinusoidal vibration while mounted in the normal operating position. The double amplitude of the vibration shall be 0.01 inch and the frequency of the vibration shall be varied smoothly from 1200 CPM to 3600 CPM and back over a time period of four minutes, this variation in frequency shall continue throughout the 2 1/2 hour period.

(e) During 1/2 hour of each 3 hour cycle, the compass system shall not be energized, nor subjected to motion or vibration.

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* 4.4.3.3.2 Test Level - At a point during the testing when any multiple of 100 test hours have been completed, each compass system shall be subjected to the following temperature tests:

(a) Low Temperature - The compass system, not energized, shall be subjected to an ambient temperature of -54° C for a period of 3 hours. The compass system shall then be energized and while at this temperature, the compass system shall be subjected to the requirements of the system scale error (free gyro) and the system scale error (slaved standard field) tests as specified in 6.9 with the limits as specified in 4.8.16.

(b) High Temperature - The compass system shall be energized and shall be subjected to an ambient temperature of 71° C for a period of 4 hours. The compass system shall then be subjected to the tests specified in 6.9 of Performance requirements as specified for the temperature tests with the limits as specified in 4.8.17.

* 4.4.3.3.3 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the tests:

(a) Whenever performance characteristics fall below the acceptance requirement (4.4.3.3.1) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

* 4.4.4 Special Tests - Special tests shall be conducted on a quantity of compass systems for the purpose of checking the effect of any design or material change on the performance of the system and to assure adequate quality control. The compass system selected for special tests may be selected from compass systems previously subjected to the sampling or reliability assurance tests.

* 4.4.4.1 Special Test Schedule - Selection of compass systems for special tests shall be made as follows:

(a) On an early compass system after every engineering or material change.

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(b) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

* 4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

* 4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken.

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(d) Where practical, include a test in the individual test to check all compass systems for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

* 4.5 Longevity Tests - Longevity tests shall be performed in accordance with Test Plan XXVIII of MIL-STD-781 except that 1000 hours operating time shall be performed on one sample of the compass system. One compass system used on the Reliability Qualification Plan may be continued until it has accumulated the required 1000 hours. Preventive maintenance may be performed during this test. Failures will be recorded and repaired, but the ratio of the equipment operating time to failures must not exceed the 200 MTBF.

* 4.5.1 Reconditioning of Longevity Test Samples - A compass system which has been subjected to the longevity test shall be reconditioned as follows:

(a) On completion of the longevity test, the compass system shall be reworked by the contractor by replacing all "wear" items, provided the cost maximum does not exceed the "cost to repair" specified in the contract (see 6.6). The "wear" items shall be determined by agreement between the contractor and the procuring activity.

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(b) After reworking,the contractor shall resubmit the compass system for acceptance.

* 4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and longevity tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Test Conditions -

4.7.1 Standard Conditions - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of 3.3.8.

4.7.2 Test Control Transformer - The test control transformer shall be a synchro with the following nominal electrical characteristics:

Input impedance 80 +J400 ohms
Rotor resistance 42 ohms
Stator resistance 10.8 ohms
Stator output voltage, line to line 11.8 volts
Accuracy (against standard) 2 minute spread

It shall have a scale from 0 to 360 degrees readable to one minute. The following procedure shall be used to calibrate the control transformer.

4.7.2.1 Designate at will and label the leads of the rotor "Hot" and "Cold".

4.7.2.2 Select one of the stator leads, designate and label it "Z" and tie this lead to the "Cold" rotor lead.

4.7.2.3 Apply 26V, 400 cycles AC to the "Hot" lead and ground the "Cold" and "Z" leads. Rotate the rotor or stator until the output of the remaining leads are at a "null" or minimum voltage, as indicated on a vacuum tube voltmeter.

4.7.2.4 Measure the voltage between the "Hot" lead and either of the remaining stator leads. If this voltage is less than the excitation voltage of

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the rotor, the synchro rotor is at "Low" null or the 270-degree position. If the voltage is greater than the rotor excitation voltage, the synchro rotor is at "High" null or the 90-degree position.

4.7.2.5 The rotor shall be placed at the "High" null or 90-degree position. A voltmeter shall then be connected across one of the remaining stator leads and the ground, with the second lead open. When the synchro rotor is turned clockwise, the voltage will increase before decreasing if the "X" stator lead is connected. The voltage will decrease before increasing if the "Y" stator lead is connected. All the leads of the synchro shall be labeled in accordance with the results of the above tests.

4.7.2.6 In use, the test control transformer may be positioned by hand or by a servo controlled motor or equivalent means.

4.7.3 Turntable - When a turntable is specified, it shall consist of a horizontal table rotatable at any uniform rate up to 30 degrees per second about a vertical pivot secured to a fixed base which carries a scale graduated from 0 to 360 degrees. Provisions shall be included that will assure the specified rates of turn and indicate the angle through which the table has been rotated from a fixed reference point.

4.7.4 Field Strength - Unless otherwise specified, the compass transmitter shall be located in a standard magnetic field that is undirectional and unvarying such as the earth's magnetic field having a horizontal field strength of 0.18 ±0.02 oersted and a vertical field strength of 0.54 ±0.02 oersted. Where a weak field is specified the horizontal components shall be 0.08 ±0.01 oersted, vertical component 0.54 ±0.02 oersted.

4.7.5 Vibration - All performance tests for which readings of the compass system are required with the exception of roll, pitch, and yaw motion shall be conducted with the directional gyro subjected to a vibration of 0.002 to 0.005 inch amplitude and a frequency of 1500 to 2000 cycles per minute along a vertical axis.

* 4.7.6 Pitch, Roll, and Yaw Motion (Scorsby) - When a pitch, roll and yaw testing apparatus is specified, it shall provide 1 1/2 through 7 1/2 degrees of angular motion either side of the vertical reference axis and operate at a speed of 5 to 7 cycles per minute and reverse uniformly once every minute.

4.7.7 Measurements - All measurements shall be made with instruments whose accuracy has been verified. If tests are conducted at the contractor's plant, the accuracy of the instruments and test equipment shall be verified periodically by the contractor to the satisfaction of the procuring activity.

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4.7.8 Attitude - Unless otherwise specified, the compass system shall be tested in its normal operating position.

* 4.7.9 Compass Transmitter - Where a compass transmitter is required it shall conform to MIL-T-19576. The vendor shall be responsible for calibration and standardization of the transmitters. A simulated compass transmitter may be used for production testing except that 10 percent of the systems shall be tested with compass transmitters supplied by the government.

* 4.7.9.1 Simulated Compass Transmitter Characteristics - The simulated compass transmitter shall be a Clifton synchro transmitter type CGH-11-B2, mounted in an indexing head and adjusted and coded for electrical zero and rotation per MIL-S-20708. The rotor shall be excited from 230 ±mv, 800 cps with R-1 "high" and R-2 "low". The high side of the 800 cps excitation voltage shall be identified with the "high" side of the 400 cps, phase A reference voltage as follows:

(a) The 800 cps excitation voltage shall pass through zero amplitude at the same instant that the 400 cps reference voltage passes through zero amplitude.

(b) The 800 cps excitation voltage shall increase negatively from zero amplitude at the same instant that the 400 cps reference voltage increases positively from zero amplitude.

4.8 Test Methods -

4.8.1 Examination of Product - Each compass system and component shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.8.2 Nutation - There shall be no spinning or excessive oscillation about the vertical axis of the directional gyro when power is first applied to the system.

* 4.8.3 Anti-Spin Brake (System B only) - Within one minute after applying power to the compass system, the anti-spin brake of the directional gyro shall operate and release the vertical gimbal.

* 4.8.4 Leveling - The gyro shall be tilted so that it is 20 degrees out of level with the horizontal. The leveling rate shall be determined to be 2 to 4 degrees per minute as the gyro levels itself. This can be done before the gyro is placed in its casing.

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* 4.8.5 Synchronizing Indicator Deflection - With the compass system operating in the slaved mode, an error signal of 1/4 degree between the compass transmitter and the system heading shall produce a visual signal at the annunciator. The compass transmitter shall be displaced 10 degrees to the left and then 10 degrees to the right of the gyro output signal. The maximum deflection of the pointer shall occur at a signal difference of 6 ±0.2 degrees.

4.8.6 Operation Selection - During this test a compass transmitter shall be connected to the MA-1 compass system and the transmitter shall be set to a heading approximately 45 degrees greater than the system heading. The mode selector switch shall be set to the compass controlled mode of operation. The latitude corrector shall be set to the latitude of the test location. The movement of the gyro shall be noted for a 15 minute period. This gyro movement shall be within the tolerance specified in Table II. The mode selector switch shall then be set to the directional gyro mode of operation and the latitude corrector shall be set to the latitude of the test location. The gyro movement shall be noted for a 30-minute period. The gyro movement shall be within the tolerance specified in Table II. The mode selector switch shall remain in the directional gyro mode of operation and the latitude corrector shall be set to the latitude of the test location, but with the hemisphere selector set to the opposite hemisphere. The gyro movement during a 30-minute period shall be within the tolerances specified in Table II.

TABLE II

MODE OF OPERATION	LATITUDE CORRECTOR SETTING	TIME (MINUTES)	GYRO MOVEMENT (DEGREES)
Compass Controlled Directional Gyro	Test Latitude	15	Between 15 and 30
Free Directional Gyro	Test Latitude opposite hemisphere	30	-15 (Sine Lat.) ±2
Free Directional Gyro	Test Latitude	30	0 ±2

* 4.8.7 Direction of Rotation - The compass system shall be operated in the compass controlled directional gyro mode. A compass repeater indicator ID-663/U in accordance with MIL-I-22075 or ID-1013/A in accordance with MIL-I-23366 shall be connected to the heading output synchro transmitter. The compass transmitter shall be set to magnetic north (0 degrees) and the system shall be synchronized. The procedures as described in Table III shall be followed. The response as indicated in Table III shall be obtained.

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TABLE III

FUNCTION	SYNCH INDICATOR MOTION OR POSITION	INDICATOR OR COMPASS CARD READING
1. (a) Rotate the compass transmitter to 30° magnetic heading	Shall indicate full scale left	0°
(b) Move the gyro clockwise until the synch indicator needle is centered	Centered	30°
2. (a) Rotate the compass transmitter to 330° magnetic heading	Shall indicate full scale right	30°
(b) Move the gyro clockwise until the synch indicator needle is centered	Centered	330°
3. (a) Rotate the compass transmitter to magnetic north (0 degrees)	Shall indicate full scale left	330°
(b) Move the gyro clockwise until the synch indicator needle is centered	Centered	0°
4. (a) Rotate manual set knob counterclockwise	Shall move to the left	Shall move in the decreasing heading direction
(b) Rotate manual set knob clockwise	Shall move to the right	Shall move in the increasing heading direction

4.8.8 Compass Control Rate - The compass system shall be set on compass controlled operation. A compass transmitter shall be set to the north magnetic heading and the system set to the 90 degrees heading. The time required for the compass system output sign:) reach an indication of 70 degrees shall be noted. This procedure shall be repeated with the compass system set to the 270 degrees heading and the time required for the compass system output signal to reach an indication of 290 degrees shall be noted. The average time for the output signal to reach the specified indications shall be not less than 10 minutes or more than 20 minutes.

4.8.9 System Scale Error (Free Gyro) - A test control transformer, as specified in 4.7.2, shall be connected to the output synchro transmitter. The latitude corrector shall be set at 90 degrees north latitude, with the system output set to north (0 degrees). Vibration shall be applied during the test. The compass system shall then be operated as a free gyro for 30 minutes and the output signal indications shall be noted. A similar procedure shall be followed and the test shall be repeated under the following conditions:

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(a) Latitude corrector set at 90 degrees south latitude; system output set at 90 degrees.

(b) Latitude corrector set at 0 degrees latitude; system output set at 180 degrees.

(c) Latitude corrector set at the latitude of the test location; system output set at 270 degrees.

The error of the output signal indication shall be calculated as follows:

$\Delta I = -7.5 (\sin L' - \sin L)$ and shall not exceed plus or minus 2 degrees in each of the above settings.

ΔI is the change of indication.

L' is the latitude set into the latitude corrector.

L is the actual latitude at the area of test.

The above test shall be repeated twice under the following conditions:

1. The gyro shall be subjected to 3 degrees reversing Scorsby motion in lieu of standard vibration. (At two of the test headings only, 90 degrees apart.)

2. The gyro shall be subjected to 15 degrees reversing Scorsby motion in lieu of standard vibration. (At two of the test headings only, 90 degrees apart.)

The system output headings of 90, 180, and 270 degrees shall be set by moving the Scorsby machine and gyro as one unit and not by moving the gyro alone or operating the set heading knob.

The difference in error between any of the readings shall not exceed two degrees.

* 4.8.10 System Scale Error (Slaved Standard and Weak Field) - The selected compass transmitter shall be mounted on a turntable in a standard magnetic field and the MA-1 compass system shall be operated in the compass controlled directional gyro mode. The compass transmitter shall be set to the magnetic north heading (0 degrees). The system shall then be set to the 10-degree heading and the gyro allowed to precess to a final heading. The system shall then be set to the 350-degree heading and the gyro allowed to precess to a final heading. The heading of the system shall be determined by the test control transformer, as described in 4.7.2, connected to the output synchro transmitter.

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The difference between the two final headings shall not exceed 0.5 degree and the average of the two readings shall be taken as the heading at this point. This procedure shall be followed at each 30-degree heading of the compass transmitter. The maximum system scale error shall not exceed 2 degrees at any heading of the system from zero to 360 degrees. This test shall be repeated with the compass transmitter subjected to a weak magnetic field (0.08 oersted). The errors of the compass transmitter shall not be included as part of the system errors.

4.8.11 Power Requirement - The compass system shall be placed in the mode of operation that consumes the most power within the condition of MIL-STD-704, Category B. The rated voltages and frequencies of power shall be applied. The AC power shall not exceed 300 VA. The difference in load between the phases shall not exceed 50 percent and the power factor shall be between 0.6 and unity. The DC power shall not exceed 40 watts.

4.8.12 Null Shifts in Slaving Amplifier - The gyro shall be replaced by a compass transmitter during this test. The MA-1 compass system shall be set to operate in the compass controlled mode. Ten minutes after the application of power, the compass system shall be synchronized. The output of the compass system shall then be monitored for a minimum of one hour. The synchronized position shall not shift by more than 0.25 degree during this time period.

* 4.8.13 Servoed and Non-Servoed Heading Synchro Transmitter Operation (System A only) - The compass system shall be operated as a free gyro. The gyro shall be mounted on a turntable capable of rotating at a uniform speed and capable of undergoing a sine wave motion around a given point. The system input for this test shall then be the motion of the turntable, either rotating at a given speed or a sine wave motion at a given frequency and amplitude. The transmitting synchro outputs shall follow smoothly a rotation input with less than 1.4 mechanical degrees error at 7 rpm. It shall also follow smoothly a sine wave input of one cycle per second and 10 degrees peak to peak amplitude with less than 20 electrical degrees phase lag. If the compass system has a non-servoed auto-pilot transmitter mounted on the gyro azimuth axis there shall be no mechanical error or electrical phase lag when subjected to the test procedures.

4.8.14 Auto-Pilot Cutout Switch - The auto-pilot cutout switch shall operate properly when the heading knob is operated.

* 4.8.15 Voltage and Frequency Variation - The varying voltages and frequency requirements shall be in accordance with MIL-STD-704, Category B. The compass system shall meet the scale error requirements of 4.8.9 and 4.8.10. The tolerances of the scale error test, 4.8.9, may be increased 0.5 degree.

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* 4.8.16 Low Temperature Operation - The compass system shall be tested in accordance with Step 2, Temperature and Altitude Test of MIL-T-5422 for 4 hours. At the end of the 4-hour period and while at the reduced temperature, the compass system shall be operated for at least one (1) hour and shall be subjected to and meet the requirements of 4.8.9 and 4.8.10. The tolerances for 4.8.9 may be increased 0.5 degree. After the compass system has returned to room temperature and has saturated for a minimum period of 4 hours the compass system shall be subjected to and meet the requirements of 4.8.9 and 4.8.10. There shall be no damage which would adversely affect subsequent operation.

* 4.8.17 High Temperature Operation - The compass system shall be tested in accordance with Step 6, Temperature and Altitude Test of MIL-T-5422. During the specified temperature the compass system shall be operated for at least one (1) hour and be subjected to and meet the requirements of 4.8.9 and 4.8.10. The tolerances for 4.8.9 may be increased 0.5 degree. After the compass system has returned to room temperature and has saturated for a minimum period of 4 hours, the compass system shall be subjected to and meet the requirements of 4.8.9 and 4.8.10. The tolerance shall remain the same. There shall be no damage which would adversely affect subsequent operation.

4.8.18 Rotation of Spin Axis through 360 Degrees - The compass system shall be operated in the directional gyro mode with the latitude corrector set to south 90 degrees. The gyro shall be subjected to a 3-degree reversing Scorsby motion. The output of the compass system shall be continuously recorded by means of a suitable recording device, for a period of 20 hours or until the gyro spin axis has rotated 360 degrees. There shall be no discontinuities or abrupt changes in the recording. The drift error during any one-hour period shall not exceed 4 degrees. The recording device shall be capable of following the system output continuously.

* 4.8.19 Vibration - The compass system shall be tested in accordance with Parts I and II, Vibration Tests of MIL-T-5422 with the following exceptions:

(a) For the (MS90317) MA-1 Compass directional gyro- The maximum double amplitude shall be increased to 0.020 inch and decreased to 1 g between 90 and 210 cps.

(b) Controller and amplifier - The maximum double amplitude shall be increased to 0.020 inch.

After completion of the vibration period the compass system shall be subjected to and meet the requirements of 4.8.9 and 4.8.10.

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4.8.20 Radio Noise Interference - The compass system shall be electrically connected to the test voltage and tested for conducted and radiated radio interference in accordance with MIL-I-6181.

4.8.21 Automatic Cutout of Compass Control and Gyro Leveling - The compass system shall be connected to a compass transmitter, synchronized and operated in the magnetically slaved mode. A signal simulating that provided by a rate of turn gyro to activate automatic cutout shall be applied to the system at the appropriate electrical connector. The compass transmitter shall be then rotated approximately 30 degrees from the synchronized position. The compass system shall not slave to this new heading. The gyro leveling circuit shall be monitored to determine that the leveling circuit is cutout at the same time that the slaving system is cutout.

* 4.8.22 Room Temperature - High Altitude - The compass system shall be tested in accordance with Step 9, Temperature and Altitude Tests, of MIL-T-5422, except the temperature shall be 25° C. While at this temperature and pressure the compass system shall be subjected to and meet the performance requirements of the System Scale Error tests 4.8.9 and 4.8.10.

* 4.8.23 Low Temperature - High Altitude - The compass system shall be tested in accordance with Step 3, Temperature and Altitude Tests, of MIL-T-5422, except the altitude shall be 50,000 feet for a period of four (4) hours. While at this temperature and pressure, the compass system shall be subjected to and meet the requirements of the System Scale Error tests 4.8.9 and 4.8.10. The tolerance for 4.8.9 may be increased 0.5 degree.

4.8.24 Shock - The compass system operating in the slaved compass controlled mode shall be tested in accordance with the Shock Test of MIL-T-5422 except the acceleration shall be 10 ±1g. Upon completion of the test, the compass system shall be subjected to and meet the requirements of the System Scale Error tests 4.8.9 and 4.8.10.

* 4.8.25 Humidity - The compass system shall be subjected to the Humidity Test of MIL-T-5422 except 71° C shall be substituted for 50° C in Step 1 and 2. At the conclusion of the test the compass system shall be subjected to and shall meet the requirements of the System Scale Error tests 4.8.9 and 4.8.10. Examination for deterioration shall be made between 24 and 48 hours after completion of the Humidity exposure. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the compass system.

* 4.8.26 Salt Spray - The Compass System shall be tested in accordance with the Salt Spray Test of MIL-T-5422. At the end of the test period the equipment shall be subjected to and meet the requirements of 4.8.9 and 4.8.10.

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* 4.8.27 Fungus Resistance - The Compass System shall be tested in accordance with the Fungus Resistance Test of MIL-T-5422, except that the test period shall be 14 days. At the conclusion of the test any evidence of the growth of fungi shall be cause for rejection, and the equipment shall be subjected to and meet the requirements of 4.8.9 and 4.8.10.

* 4.8.28 Reconditioning of Tested Equipment- Compass Systems which have been subjected to acceptance and longevity tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor may resubmit the Compass System for acceptance.

* 4.8.29 Presubmission Testing - No item, part or complete Compass System shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8.30 Rejection and Retest - Compass Systems which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging - Each component of the Compass System shall be packaged in accordance with MIL-STD-794, Level A or C as specified in the contract or order (see 6.6). For Level A packaging, the method of preservation shall be in accordance with MIL-P-116, Method IC, utilizing a rectangular container. The container shall conform to MIL-C-4150.

* 5.2 Packing - Each component of the Compass System shall be packed in accordance with MIL-STD-794, Level A, B, or C as specified in the contract or order (see 6.6). The package and exterior shipping container may be one and the same container.

* 5.3 Rough Handling Tests - Rough handling tests shall apply to all levels of packaging and packing. The tests performed on a prototype example shall be Rough Handling, Dummy Load and Preproduction Tests of MIL-STD-794. No vibration tests are required. Two pilot shipping containers are required to be tested. The heights of drop shall be 30 inches for the free-fall drop and 36 inches for the free-fall cornerwise drop. All testing shall be done on all axes and in each direction in each axis and with actual equipment as packaged for shipment. After tests, the compass systems shall not suffer damage or subsequently fail to provide performance as specified. All testing shall be done at the Contractor's plant. The contractor

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shall notify the procuring activity in sufficient time in advance of prototype testing to permit Government representation at the test site.

* 5.4 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-129 and as follows:

* 5.4.1 Precautionary Marking - The following precautionary markings warning of the fragility of the item shall appear on two opposite sides of each interior package and shipping container. Containers shall be marked with arrows on two opposite sides of each interior package and shipping container to indicate which surface must be always up:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE
UP



In addition, containers shall be marked "Reusable, OPEN WITH CARE" in letters at least 2 inches high, and on two opposite sides of each container.

6. NOTES

6.1 Intended Use - The Compass System covered by this specification is intended for use in fighter, attack, patrol, transport and cargo type aircraft whose range of operation will be world wide including polar regions as well as all aircraft requiring medium precision heading reference stabilization. This system will provide directional information to remote indicators, radar and navigational equipment and furnish directional control for autopilots, bomb directors, automatic approach equipment and other auxiliary systems.

* 6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 V DC and 115/200 ±1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

* 6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight

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or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

* 6.4 As a general rule nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the nonrepairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

* 6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

* 6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1).
(c) Cost to repair (see 4.5.1).
(d) Supply compass transmitter as required.

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(e) Designate Government testing facility for assigned testing.
(f) Supply designated model in accordance with requirement of MIL-E-5400.

6.7 Asterisk - In specification revisions and superseding amendments an "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the notations and relationship to the last previous issue.

6.8 Items of Equipment - The compass system covered in this specification shall be classified as Type MA-1, and shall consist of a directional gyro, controller, and amplifier. The number of components may be decreased upon authorization by the procuring activity.

Component		Military Standard	
		System A	System B
Amplifier,	MA-1 Compass	MS90315	MS90318
Controller,	MA-1 Compass	MS90316	MS90319
Directional gyro,	MA-1 Compass	MS90317	MS90320

* 6.8.1 Associated Equipment - The compass system shall operate with the following associated equipment:

MIL-I-19576 Transmitter, Remote Compass
MIL-I-22075 Indicator, Bearing Distance Heading
MIL-I-23366 Indicator, Horizontal Situation

* 6.9 Performance Characteristics to be Measured - The system scale error (free gyro) test and the system scale error (slaved standard field) test shall be conducted at the following times:

(a) Upon receipt at the designated laboratory or test facility.

(b) Upon completing 2, 5, 10, 16, and 20 operating cycles.

(c) Upon completing every 8 cycles from 24 to 96 inclusive.

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(d) Upon completing every 16 cycles from the 112 cycle and thereafter.

(e) During the temperature exposure tests.

A linear increase in free gyro drift from 4 degrees per hour at 500 hours to 8 degrees per hour at 1000 hours shall be allowed for any system when the operating time of each component of the system exceeds 500 hours.

The system output headings of 90 degrees, 180 degrees, and 270 degrees shall be set by moving the gyro and not by moving the heading set knob.

* 6.10 Applicable Documents -

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-C-4150	Case, Carrying, Water-Vaporproof
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-C-5015	Connectors, Electric, AN Type
MIL-W-5088	Wiring: Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-007788(Wep)	Plates, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing

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Military (continued)

MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-19576	Transmitter; Remote Compass, Thin Wing Type ML-1 (Unstabilized)
MIL-S-20708	Synchros, 60 and 400 Cycle, General Specification for
MIL-I-22075	Indicator, Bearing Distance Heading
MIL-I-23366	Indicator, Horizontal Situation
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS17322	Meter, Time Totalizing Miniature, Digital 115 volt 400 Cycle
MS90315	Amplifier, MA-1 Compass (System A)
MS90316	Controller, MA-1 Compass (System A)

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Military (continued)

MS90317	Directional Gyro, MA-1 Compass (System A)
MS90318	Amplifier, MA-1 Compass (System B)
MS90319	Controller, MA-1 Compass (System B)
MS90320	Directional Gyro, MA-1 Compass (System B)

6.10.1 Availability of Documents - When requesting specifications standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 1051, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.11 Definitions -

6.11.1 Hermetic Seal - A hermetic seal shall be a perfectly closed and airtight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals which are dependent upon gaskets.

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c. Reason/Rationale for Recommendation:

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